

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

SKILLS

- Languages** – Python, Go (Golang), SQL, C, C++.
- Backend & APIs** – FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio, Design Patterns, Service-Oriented Architecture (SOA).
- AI & Agents** – LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, Milvus, pgvector.
- Data & Streaming** – Kafka, Flink, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- Cloud & Infra** – GCP, AWS, Docker, Kubernetes, Linux, CI/CD, Bazel, Observability (ELK, New Relic, Datadog, PostHog, LangSmith).
- Core** – Distributed Systems, Microservices, System Design (HLD/LLD), Multithreading, Concurrency, Caching, Sharding, Mentorship.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building **AssortSmart** so retailers decide **what to buy, how much, and for which stores** with a planner chat (**FastAPI, LangGraph, MCP**) and shared **Go (Gin)** write APIs, targeting plan turnaround from days toward **under 1 hour** and search from **1 to ≥ 20 configs/plan**.
- Cutting measured clustering-run failures from **8.5% (37/437 on kik)** toward **under 2%** (target) with **14** audited read-only tools and **3** human confirm steps so the agent never writes SQL.
- Replacing a projected **12B**-row store-week design with a **$\sim 25\text{M}$** -row aggregate so month rollups stay **sub-second** and cell edits stay **$\sim 0.4\text{ ms}$** (measured) without exploding the table.
- Adopting per-tenant **ClickHouse (63 tables / 8 layers)**, insert-only, refresh by swapping partitions, agent read-only) as the planning store after a same-rows POC cut **250M**-row pivots from **189s to 12.3s** ($\sim 15.5\text{x}$, measured).
- Tech** – Go, Gin, Python, FastAPI, LangGraph, MCP, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

Software Development Engineer 2

UBER (via EPAM Systems), Bangalore

July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned design and architecture for Uber's **FRM Risk Scoping platform (FastAPI, MySQL, React)** across **8 screens and 30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70% cut** in manual reconciliation (**~ 2 weeks to ~ 3 –4 days**).
- Designed an **11-table MySQL** schema (SQLAlchemy 2.0) with clear handler / service / repository layers that auto-flags material items across **55 line items and 14 entities**.
- Owined the Sheets to **MySQL** recon v2 migration across **18 files**, checking HFM extracts against public **10-Q** filings and fixing a join bug where two columns shared the same name.
- Led 3 engineers** through design reviews, API contracts, and CI quality gates for releases feeding PwC audit work papers.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built a menu pipeline (**Selenium $\rightarrow$ Kafka $\rightarrow$ Flink**) that brought onboarding from **24h to 2h** and saved **\$600K+ annually** on **30K+ menus/month**.
- Built a **LangChain RAG + Gemini 2.5 Pro** extractor on **Milvus** that turns multilingual **PDF/image** menus into Uber Eats schema at **98% fidelity / 100% schema consistency** (offline eval).
- Raised successful ingestions to **95%+** with IP rotation, proxy pools, and retries when sites block scrapers.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

ANZ Driver Document Compliance (Uber Mobility)

- Automated Uber driver and vehicle document checks for the **ANZ** region, achieving **99.9%** compliance with local authorities and eliminating about **20 hours/week** of manual verification.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Migrated a PHP (**Laravel**) monolith to **FastAPI microservices** step by step for **1,500+ clients**, bringing p95 from **1.2s to 300ms** while mentoring **2 engineers**.
- Built a bulk e-invoice path on **Kafka** and **PostgreSQL** (tables split by tax quarter) that processed **1M+ government submissions/day** and **100K+/import**, lifting throughput from **700 to 4,000 requests/min**.
- Made IRP retries safe with **idempotency keys, retries, and a dead-letter queue**, and used Redis caching to lower repeat DB reads by **30%**.
- Set up **ELK and New Relic** alerting plus a client usage dashboard that lowered incident triage **70%** and support tickets **35%**, raising coverage **35% to 82%** at **98%** deploy success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Stabilized doubt-support for **10K+ daily queries** and **10x** contest spikes by migrating the backend from PHP to **Django (Python)**.
- Lifted premium subscriptions **15–20%** by shipping voting, pinning, and locking REST APIs on MySQL, MongoDB, Redis, and Elasticsearch.
- Raised course sales **30%** by launching an influencer earnings and analytics dashboard (transactions, coupons, filters).
- Raised operational efficiency **70%** with scheduled jobs for video processing, reminders, and recording cleanup.

EDUCATION

B.Tech (Information Technology)

Institute of Engineering & Technology, Lucknow

July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).